

Audit Research

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Audit Project

The scope of this project is to devise, implement, and evaluate a system that performs audit log parsing, (possibly) reduction, and generation of a graph (or similar structure) that captures the interactions between the events in audit logs.

Objectives

1. Parse audit logs, extracting relevant information and discarding redundant and non-informative data
2. Store the parsed data in a structured format that captures:
 - the relationships between the events and the entities involved
 - the temporal dimension of the events
 - the context in which the events occurred
3. Evaluate the performance of the system in terms of:
 - the correctness of the generated structure
 - we should investigate the loss of information and the choices that must be made between preserving data and discarding it
 - the effectiveness of the structured data
 - the information that can be extracted
 - the ease-of-use of the data structure and the preservation of causality relationships
 - the efficiency of the parsing process, e.g.
 - the spatial and temporal complexity in traversing the data
 - the time required to parse a log file
 - the memory footprint of the system
 - its feasibility as a real-time system (not necessarily)

Timeline and milestones

(V) = Valentino, (M) = Matteo

Weeks 1-2 (May 7-20)

- (M+V) Literature review
 - (audit) log parsing and analysis
 - graph-based data structures
 - data provenance
- (M+V) Initial investigation of K8s audit logs

- Set up infrastructure to collect logs
- Identify priority events to parse
 - * Assess possible problematic areas
- Results
 - A literature review (shared) document
 - A shared cluster with K8s running to collect logs from
 - Initial rough idea of events to focus on

Weeks 3-7 (May 21-June 17)

- (M+V) Designing the system
 - Choosing which data structure to use
 - Outlining a definitive priority list of events to parse
 - * This also includes which Service Accounts to focus on first
 - Understanding how to include the temporal dimension
- (V) Initial implementation of parser for logs
 - Parse JSON logs
 - Extract relevant information
 - Build a data structure to store the parsed data
 - * Optimize for fast access and traversal
- (M) Work on understanding the relationships
 - Understand: when I do X, what is generated? Who is involved?
 - Which events are related to each other? Which can be discarded?
- Results
 - A shared document with the design of the system
 - An ordered list of *milestones* to reach, w.r.t. the capabilities of the system, along with the difficulties in doing so

Weeks 8-10 (June 18-July 1)

- Continuation of the work from the previous weeks, step by step adding more features to the parser making it more capable of extracting information from the logs.
- Results
 - An early implementation of the parser that should already be able to extract some information, albeit not programmatically.

Weeks 11-14 (July 2-29)

- (V) Focus on: querying the data structure
 - An user should be able to quickly find the information they are looking for.
 - * e.g. I put in a user and I can "follow" what the user did
 - * this possibly creates subgraphs that are "isolated" from the rest of the graph, on the fly
- (M) Initial assessment of the system

- Correctness of the extracted information
- Effectiveness of the structured data and its usability
- Efficiency of the parsing process
- In reality this should be a continuous process
- Results
 - A prototype of a system that is now queryable
 - A document outlining the evaluation process and the results obtained

Weeks 15-18 (July 30-August 26)

- (V) Refining queries
 - Could be the right moment to start making *user stories* to understand how the system could be used and what features users would like to see
 - Strong focus on the temporal dimension
 - * Assume an incident happened at time X, what happened in timespan $[X - 1h, X + 1h]$?
 - * Is it possible to "replay" the events that led to the incident?
- (M) Work towards feature-completeness
 - The system should be able to parse all the events we set out to parse
 - Each event should be thoroughly tested, the choices made strongly justified and documented
- Did we meet the list of milestones we set up?

Yes Perfect, we can move on to the next phase

No Take a time machine and go back to the previous weeks to fix the issues

Weeks 19-20 (August 27-September 9)

- (M+V) Starting the publication draft
 - The document should generalize the work done, making it suitable for not just Kubernetes logs but for any kind of audit log (e.g. AWS, GCP)
 - Still to be decided which type of publication we are aiming for
- (M+V) Final assessment of the system
 - The system should be able to parse logs in a reasonable amount of time, and the data structure should be efficient in storing the data
 - The system should be able to answer queries in a reasonable amount of time
 - The query system should be user-friendly, and the results should be easily interpretable
 - The system could have some kind of visualization to help users understand the data
- If something went wrong, this is the correct moment to apply fixes and improvements to the system.
- Results
 - A first WIP of the publication, still in the early stages of writing
 - A final assessment of the system
 - A visualization of the data structure
 - A final report on the project

Weeks 21-27 (September 10-October 28)

- Writing the publication
- Finalizing the system
- Closure work
- Results
 - A publication
 - A final report on the project
 - A final version of the system

Misc notes

- Accessing secrets using printenv does not leave a trace in the audit logs
- (to be expanded)